0:01

1 second

Let's do some calcium. Okay. Calcium.

0:10

10 seconds

I don't know. I feel like um some students it it's like the things that the students that I've noticed that they

0:19

19 seconds

struggle with um in terms of electrolytes is sodium and remembering its relationship with fluid

0:28

28 seconds

and um like all the different kinds of fluids and the hypotonic and the hypertonic blah blah blah. Okay, that and the calcium. Okay.

0:39

39 seconds

And I feel like there's just like one specific thing in calcium that um

0:48

48 seconds

I don't know, it just always trips students up. But anyways, let's talk about calcium. Normal calcium is 8.4 to

0:54

54 seconds

10.5. Okay? And I'm giving you little hints. So N for sodium, N for neuro.

1:02

1 minute, 2 seconds

Okay? Potassium heart. Calcium is bones and muscles.

1:09

1 minute, 9 seconds

Okay, let's go down here.

1:16

1 minute, 16 seconds

We're going to do hypocalcemia less than 8.4 and hyper hypercalcemia greater than 10.5. All right.

1:28

1 minute, 28 seconds

Uh this one I'm going to start with hypocalcemia because hypocalcemia

1:39

1 minute, 39 seconds

well I'll tell you what the signs and symptoms but um hypocalcemia can be caused by

1:46

1 minute, 46 seconds

low vitamin D um and subsequently like CKD. Okay,

1:54

1 minute, 54 seconds

those kind of go together. Now the vitamin D isn't always related to CKD.

1:58

1 minute, 58 seconds

We can just have a poor intake of vitamin D. But the relationship with vitamin D and calcium is that we need

2:05

2 minutes, 5 seconds

the vitamin D in order to absorb the calcium. Okay?

2:10

2 minutes, 10 seconds

So vitamin D is very important for calcium absorption. If we don't have the vitamin D, we can't absorb calcium. We have low calcium. And the kidneys actually work to activate the vitamin D.

2:19

2 minutes, 19 seconds

And so if our kidneys aren't working, then we essentially have low vitamin D and low calcium. Okay, great. So, that's kind of how those fit together.

2:28

2 minutes, 28 seconds

Um, diarrhea is a weird one. I hate the GI like the vague GI things, man. It's like I God, I can never remember like

2:36

2 minutes, 36 seconds

outside of the metabolic alkalossis and the metabolic acidosis with your diarrhea. Like that makes sense to me, but these electrolyte imbalances, I cannot keep track with the diarrhea.

2:47

2 minutes, 47 seconds

I Okay, [gasps] anyways, random diarrhea.

2:52

2 minutes, 52 seconds

But I think that the thing that gets students the most, I don't know, it trips them up the most

3:00

3 minutes

is when we start talking about parathyroid hormone. So low calcium can be due to the low parathyroid hormone.

3:08

3 minutes, 8 seconds

And if we remember from anatomy and physiology what the parathyroid hormone does.

3:15

3 minutes, 15 seconds

Um our parathyroid hormone is important in calcium regulation. But I like to just simplify this and say that

3:23

3 minutes, 23 seconds

parathyroid hormone sorry I don't really want to pay attention to this arrow. Low parathyroid hormone will call cause hypocalcemia but I'm just talking parathyroid hormone just in general.

3:33

3 minutes, 33 seconds

Okay. So parathyroid hormone breaks down bones. So parathyroid hormone normally gets released

3:42

3 minutes, 42 seconds

when we have uh low calcium. Okay. If we have now this is normal. If we have low

3:49

3 minutes, 49 seconds

calcium I have parathyroid hormone that increases. It breaks down my bones. I

3:57

3 minutes, 57 seconds

think of it just is oh my god that's a terrible bone. Okay. I just think of it like parathyroid hormone is like um

4:06

4 minutes, 6 seconds

I don't even know what I'm like trying to describe but like something like chipping away at the bone. It's just like ting ting ting ting ting ting ting

4:14

4 minutes, 14 seconds

on that bone. And the calcium is just like cuz we store calcium in our bones.

4:19

4 minutes, 19 seconds

So all that calcium that's found in our bone when our parathyroid hormone breaks it down is then going into our bloodstream to

4:28

4 minutes, 28 seconds

raise our calcium level. Okay, that's how it normally works. So, if I have low parathyroid hormone, that means I'm not

4:36

4 minutes, 36 seconds

breaking down my bones and I'm not getting the calcium in my blood.

4:41

4 minutes, 41 seconds

And the main reasons that we're getting low parathyroid hormone is from like pro like neck surgeries. Sorry, I was want

4:50

4 minutes, 50 seconds

to say a lot of things there. Okay, neck surgeries or thyroid surgeries like a thyroidctomy. I'm taking out my thyroid.

4:57

4 minutes, 57 seconds

They might take out part of the thyroid and the thy the oh my gosh they might take out part of the parathyroid

5:04

5 minutes, 4 seconds

and so the parathyroid will not work because the parathyroid is right next to the thyroid. Anyways those are your main

5:11

5 minutes, 11 seconds

causes of hypocalcemia. Okay remember your vitamin D CKD relationship and parathyroid hormone. Okay the

5:20

5 minutes, 20 seconds

diarrhea is the random whatever. Okay signs and symptoms here. Okay, for your signs and symptoms, your hypocalcemic

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5 minutes, 27 seconds

patient I think looks, if you guys know Seinfeld looks like Kramer from Seinfeld. Okay, anytime you see the word

5:35

5 minutes, 35 seconds

spasms, tetany or prolonged muscle contractions, um,

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5 minutes, 42 seconds

cramps, that's I don't know. I think Kramer from Seinfeld. Okay, he's just spazzy.

5:54

5 minutes, 54 seconds

So, I like to think that Kramer has hypocalcemia. Okay, the other weird stuff with the hypocalcemia, and we already talked

6:02

6 minutes, 2 seconds

about it when we talked about alkyossis, is the numb tingly mouth and fingers

6:09

6 minutes, 9 seconds

because calcium is important for muscle function or function. Okay, so um if I don't have enough of it, they

6:17

6 minutes, 17 seconds

can't function properly. I get this numbness and tingly tingliness. tingling.

6:24

6 minutes, 24 seconds

Um, confusion. Uh, confusion is so vague, you guys. Gross. Okay. Lethargy.

6:30

6 minutes, 30 seconds

Lethargy is vague. But these patients are also at risk for seizures.

6:36

6 minutes, 36 seconds

Um, dysriythmias also.

6:41

6 minutes, 41 seconds

So, I feel like if we just remember muscles and bones, it kind of gets the gist. And Kramer from Seinfeld. Yeah.

6:48

6 minutes, 48 seconds

Okay. Now there are two very specific kinds of um spasms that we're looking

6:56

6 minutes, 56 seconds

for in hypocalcemia and this is shiovastex and tricos sign. These are like quintessential enclelex things you guys.

7:08

7 minutes, 8 seconds

Okay. Shastex and trisos. So shiovastc sign a positive shiovastc sign is if I

7:17

7 minutes, 17 seconds

put a blood pressure cuff on someone's arm with hypocalcemia and I inflate that cuff higher than

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7 minutes, 26 seconds

their systolic blood pressure and I let it sit there for your slides say three minutes. I feel

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7 minutes, 33 seconds

like that's a really long time. Anyways, that's how long it takes for this to work or to see this. Their hand will

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7 minutes, 39 seconds

start to kind of spasm and their um there's a picture of it in your slide. I can't really describe it. Their hand

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7 minutes, 47 seconds

their wrist will kind of uh flex and they kind of like more cup their hand and it's a very specific sign.

7:56

7 minutes, 56 seconds

Okay, this is a spasm.

7:59

7 minutes, 59 seconds

Triso sign is Um, oh my gosh. You know, this is really embarrassing because did I just mix those up? Oh my gosh.

8:11

8 minutes, 11 seconds

I just need to fact check myself really quick. Sometimes when I do these and I'm like, "Oh my gosh, am I even telling you

8:18

8 minutes, 18 seconds

the right thing?" Oh, I'm so glad I did.

8:26

8 minutes, 26 seconds

Okay. [laughter] No, I just described you Tresos. Oh my gosh. Sorry that's just the blood pressure.

8:34

8 minutes, 34 seconds

Shvastex is the face. Okay. So if we like touch where the facial nerve intervates, okay, if you just kind of

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8 minutes, 43 seconds

like brush the side of their cheek, their face will twitch.

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8 minutes, 49 seconds

Okay. And uh that's shvast sign the face. Trisos is the blood pressure thing. Whatever.

8:58

8 minutes, 58 seconds

Sorry.

8:59

8 minutes, 59 seconds

very classic hypocalcemic signs. Okay, so they're spasmy, confused, risk for seizures, risk for dysriythmias.

9:09

9 minutes, 9 seconds

So how are we going to treat our hypocalcemic patients? Well, we can obviously replace calcium, but we also

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9 minutes, 17 seconds

need to think about our vitamin D, especially if we have a vitamin D issue.

9:21

9 minutes, 21 seconds

Okay, we can give IV calcium gluconate if their calcium's really low. And I

9:28

9 minutes, 28 seconds

want to give anytime I'm pushing electrolyte. So I never push potassium.

9:33

9 minutes, 33 seconds

Sodium we have to change slowly either going from like a hypo situation back up to normal or we're bringing down a

9:40

9 minutes, 40 seconds

hyperetreia. It has to go slow. If I push calcium or magnesium, it has to be

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9 minutes, 47 seconds

slow. Okay. Um, these can be a calcium and mag can be given IV push. However,

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9 minutes, 55 seconds

it needs to be extremely slow. I'm talking like a 10-minute IV push or more. Okay, it's kind I don't know, man.

10:04

10 minutes, 4 seconds

It's kind of silly that we even push it.

10:06

10 minutes, 6 seconds

I mean, it's like 10 minutes. My god, just put on a pump. Anyways, whatever.

10:10

10 minutes, 10 seconds

I'm not a pharmacist. I don't make the meds, but you can give it IV push just very slowly. Okay, so that's our main

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10 minutes, 18 seconds

treatment. We're just replacing their potassium. Okay. High potassium foods, orange juice, green leafes, dairy

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10 minutes, 25 seconds

products. Great. Nursing stuff. Here we go. Well, they're at risk for seizures. So, seizure precautions.

10:34

10 minutes, 34 seconds

Okay. A neuromuscular assessment.

10:38

10 minutes, 38 seconds

Assess for your chiovastcex and your tricos.

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10 minutes, 41 seconds

Check deep tendon reflexes. Okay. Are they spasming? Do they have the reflexes still?

10:48

10 minutes, 48 seconds

And we're going to check their respiratory status, their respiratory rate, and their any like adventitious breath sounds.

10:56

10 minutes, 56 seconds

The reason for this is because these spasms can happen in any of their

11:03

11 minutes, 3 seconds

muscles. And one of the muscles that it can happen in is our muscles like the smooth muscle that um surrounds our

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11 minutes, 11 seconds

airways. And so we can get tightening and spasms of these muscles that surround my airway causing like a bronco constriction. And so one of the things

11:18

11 minutes, 18 seconds

that we're going to possibly see in a patient that's very hypocalcemic is something called strider. And hopefully you guys talked about this in

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11 minutes, 26 seconds

health assessment. Um but that is like indicative of like an upper airway problem

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11 minutes, 33 seconds

when they have strider and upper airway narrowing. Okay.

11:38

11 minutes, 38 seconds

Um so those are the things that we're going to be looking for there. Um, let's just take this a step further because it's me and I'm sorry, but our

11:46

11 minutes, 46 seconds

hypocalcemic patients, if they're low on calcium, they're also probably at risk for what problem? And if you said

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11 minutes, 53 seconds

fractures and osteopenia, you are absolutely correct. Okay. So, if we don't have calcium to build strong bones,

12:01

12 minutes, 1 second

um, then our bones are going to be weak and brittle. Okay? So, they're more likely to break and um the bone mass

12:08

12 minutes, 8 seconds

density is going to be very low. Okay, so that's hypocalcemia. Kramer and Seinfeld. There we go. Hypercalemia.

12:17

12 minutes, 17 seconds

Okay, if low parathyroid hormone causes hypocalcemia, then high parathyroid hormone causes hypercalcemia.

12:25

12 minutes, 25 seconds

Parathyroid hormone acts to break down the bones. If I'm breaking down the bones, I'm releasing more calcium into my bloodstream, thus hypercalcemia.

12:34

12 minutes, 34 seconds

Pathological um fractures or bone cancers. Okay. Will cause that bone to

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12 minutes, 41 seconds

break down and release um calcium into the blood. Vitamin D toxicity. Weird but

12:48

12 minutes, 48 seconds

can happen. And prolonged bed rest. Okay. Prolonged periods of

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12 minutes, 55 seconds

immobility. We have very quick muscle breakdown and we have bone breakdown. So think of these as like aka bone

13:04

13 minutes, 4 seconds

breakdown. Anything that's going to cause my bones to break down will cause potentially hypercalcemia.

13:09

13 minutes, 9 seconds

Okay. Your signs and symptoms associated with your hypercalcemia. I like to just remember this nursing great thing. Bones, stones, and groans.

13:20

13 minutes, 20 seconds

Okay. They are going to have pain. They can have abdominal pain. They can have stones. So, kidney stones and pain from that.

13:31

13 minutes, 31 seconds

Um, their bones may hurt because they're breaking down.

13:36

13 minutes, 36 seconds

They might be thirsty. They have this vague muscle weakness that we know and love and we see in almost every electrolyte imbalance [laughter]

13:45

13 minutes, 45 seconds

and constipation. I remember because of the groans. Like that's not comfy. They get constipated.

13:53

13 minutes, 53 seconds

Okay.

13:55

13 minutes, 55 seconds

So those bone stones and groans for your hypercalcemic patient treatment. All right. There's an inverse

14:03

14 minutes, 3 seconds

relationship with calcium and phosphorus.

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14 minutes, 6 seconds

So if I have high calcium I would like to give some foss to help drive that calcium down. Okay.

14:20

14 minutes, 20 seconds

High foss would lead to low calcium.

14:24

14 minutes, 24 seconds

Okay. So, that's kind of the rationale there. And then I would want to give some IV fluids to help um kind of flush

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14 minutes, 33 seconds

out that calcium. I could give IV fluids and loop diuretics. Okay. Weird concept. I'm diarasing them.

14:42

14 minutes, 42 seconds

I'm giving off fluids, but then I'm giving them more fluids. What? Okay. The whole idea of this, if you see this ordered, it's very odd. Um, and it still

14:52

14 minutes, 52 seconds

catches me off guard sometimes. I'm like, what are we doing? Okay, if we give somebody a diuretic with IV fluids, it's to help the

15:01

15 minutes, 1 second

excretion of whatever we're trying to get out. We're trying to flush the system of something. And so, we want to hit them with more fluids to try to

15:09

15 minutes, 9 seconds

dilute it, but also a diuretic to actually get rid of it. So, whatever.

15:16

15 minutes, 16 seconds

Calcetonin. Calcetonin is the hormone we can give like synthetic calcetonin um to drive down the calcium levels.

15:24

15 minutes, 24 seconds

Calcetonin and parathyroid hormone work opposite of each other. Okay. If it's really bad, we can do diialysis

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15 minutes, 34 seconds

and a very simple treatment for these patients especially that have like prolonged bed rest and immobility. Well,

15:42

15 minutes, 42 seconds

when we start to use that extremity again or you know the bones that have been immobilized

15:49

15 minutes, 49 seconds

[sighs]

15:49

15 minutes, 49 seconds

like let's obviously we're talking about legs here with ambulation but um just using the muscles and stuff will put the

15:57

15 minutes, 57 seconds

calcium back in the bones and drive that calcium level down. Okay,

16:03

16 minutes, 3 seconds

so nursing stuff try to flush out that extra calcium.

16:09

16 minutes, 9 seconds

Oh my gosh. I mean, whatever. It's fine. Monitor intake and output.

16:17

16 minutes, 17 seconds

Um, everyone's on fall precautions. That doesn't matter. Okay. Assess their pain.

16:23

16 minutes, 23 seconds

We want to tell them about a low calcium diet and put them on a cardiac monitor because of the risk for dysriythmias.

16:29

16 minutes, 29 seconds

Okay. Anytime there's a risk for dysriythmias, cardiac monitor. All right.